

Лабораторная работа №1

UNIX знакомство: useradd, nano, chmod, docker, GIT, CI, CD

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Цель

Познакомиться с основами администрирования программных комплексов в ОС семейства UNIX, изучить принципы виртуализации и контейнеризации, освоить работу с системой контроля версий Git.

Задачи и выполнение

1. Работа в ОС (Windows + WSL или виртуальная машина)

Для выполнения лабораторной работы на Windows использовалась подсистема WSL2 с установленным дистрибутивом Ubuntu 22.04.

1. В каталоге `/usr/local/` создать две директории: `folder_max`, `folder_min`.
2. Создать две группы пользователей: `group_max`, `group_min`.
3. Создать двух пользователей: `user_max`, `user_min`, добавив их в соответствующие группы.
4. Настроить права доступа:

```
sudo chown -R user_max:group_max /usr/local/folder_max
sudo chown -R user_min:group_min /usr/local/folder_min
```

```

katya@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ sudo mkdir -p /usr/local/folder_max /usr/local/folder_min
[sudo] password for katya:
katya@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ sudo groupadd group_max
katya@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ sudo groupadd group_min
katya@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ sudo useradd -m -G group_max user_max
katya@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ sudo useradd -m -G group_min user_min
katya@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ sudo passwd user_max
New password:
Retype new password:
passwd: password updated successfully
katya@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ sudo passwd user_min
New password:
Retype new password:
passwd: password updated successfully
katya@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ sudo chown -R user_max:group_max /usr/local/folder_max
katya@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ sudo chmod 770 /usr/local/folder_max
katya@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ sudo chown -R user_min:group_min /usr/local/folder_min
katya@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ sudo chmod 770 /usr/local/folder_min
katya@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ sudo setfacl -m g:group_max:rxw /usr/local/folder_MIN
sudo: setfacl: command not found

```

Рис. 1:

5. Создать и выполнить скрипт от имени `user_max` в `folder_max`, который пишет текущую дату/время в `output.log` в той же директории.

```

user_max@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ nano /usr/local/folder_max/script.sh
user_max@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ ./script.sh
bash: ./script.sh: No such file or directory
user_max@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ cat /usr/local/folder_max/output.log
cat: /usr/local/folder_max/output.log: No such file or directory
user_max@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ cd /usr/local/folder_max
user_max@LAPTOP-MK7SQ0DI:/usr/local/folder_max$ ./script.sh
user_max@LAPTOP-MK7SQ0DI:/usr/local/folder_max$ cat /usr/local/folder_max/output.log
Fri Mar 20 18:32:36 MSK 2026
user_max@LAPTOP-MK7SQ0DI:/usr/local/folder_max$ exit

```

Рис. 2: script1

6. Создать и выполнить скрипт от имени `user_max` в `folder_max`, который пишет дату/время в `output.log` в `folder_min`.

```

katya@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ sudo -u user_max bash
user_max@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ echo 'date >> /usr/local/folder_min/output.log' >> /usr/local/folder_max/script_to_min.sh
user_max@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ chmod +x /usr/local/folder_max/script_to_min.sh
user_max@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ /usr/local/folder_max/script_to_min.sh
user_max@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ cat /usr/local/folder_min/output.log
Fri Mar 20 18:57:49 MSK 2026
user_max@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ |

```

Рис. 3: script folder max > folder min

7. Выполнить от имени `user_min` скрипт в `folder_max`, который пишет дату/время в `output.log` в `folder_min`. Доступ должен быть разрешён через ACL.

```

katya@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ sudo -u user_min bash
user_min@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ /usr/local/folder_max/script_to_min.sh
bash: /usr/local/folder_max/script_to_min.sh: Permission denied
user_min@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ |

```

Рис. 4: script folder max > folder min

8. Создать и выполнить скрипт от имени `user_min` в `folder_min`, который пишет дату/время в `output.log` в `folder_max`. Доступ должен быть закрыт — ожидается отказ.

```

katya@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ sudo -u user_min bash
[sudo] password for katya:
user_min@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ echo '#!/bin/bash' > /usr/local/folder_min/script_to_max.sh
user_min@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ echo 'date >> /usr/local/folder_max/output.log' >> /usr/local/folder_min/script_to_max.sh
user_min@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ chmod +x /usr/local/folder_min/script_to_max.sh
user_min@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ /usr/local/folder_min/script_to_max.sh
/usr/local/folder_min/script_to_max.sh: line 2: /usr/local/folder_max/output.log: Permission denied
user_min@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ |

```

Рис. 5: script $folder_{min} > folder_{max}$

9. Вывести перечень прав доступа для папок и их содержимого.

```

user_min@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ ls -ld /usr/local/folder_max
drwxrwx--- 2 user_max group_max 4096 Mar 20 18:48 /usr/local/folder_max
user_min@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ getfacl /usr/local/folder_max
getfacl: Removing leading '/' from absolute path names
# file: usr/local/folder_max
# owner: user_max
# group: group_max
user::rwx
group::rwx
other::---

user_min@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ getfacl /usr/local/folder_min
getfacl: Removing leading '/' from absolute path names
# file: usr/local/folder_min
# owner: user_min
# group: group_min
user::rwx
group::rwx
group:group_max:rwx
mask::rwx
other::---

user_min@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ ls -la /usr/local/folder_max
ls: cannot open directory '/usr/local/folder_max': Permission denied
user_min@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ ls -la /usr/local/folder_min
total 16
drwxrwx---+ 2 user_min group_min 4096 Mar 20 22:10 .
drwxr-xr-x 12 root      root      4096 Mar 20 18:04 ..
-rw-rw-r-- 1 user_max user_max   29 Mar 20 18:57 output.log
-rwxrwxr-x 1 user_min user_min   53 Mar 20 22:11 script_to_max.sh
user_min@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ exit
exit
katya@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ sudo -u user_max bash
user_max@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ ls -la /usr/local/folder_max
total 20
drwxrwx--- 2 user_max group_max 4096 Mar 20 18:48 .
drwxr-xr-x 12 root      root      4096 Mar 20 18:04 ..
-rw-rw-r-- 1 user_max user_max   29 Mar 20 18:32 output.log
-rwxrwxr-x 1 user_max user_max   53 Mar 20 18:30 script.sh
-rwxrwxr-x 1 user_max user_max   53 Mar 20 18:56 script_to_min.sh
user_max@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ |

```

Рис. 6: Права и содержимое

2. Контейнеризация (Docker)

1. Создать скрипт `date_script.sh`, который пишет текущую дату и время в файл `output.log` в текущей директории.
2. Создать `Dockerfile`, который:
 - использует базовый образ `ubuntu:22.04`;
 - устанавливает пакет `nano`;
 - копирует скрипт в контейнер;
 - задаёт команду по умолчанию.

```
katya@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ docker --version
Docker version 28.2.2, build 28.2.2-0ubuntu1~24.04.1
katya@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ echo '#!/bin/bash' > ~/date_script.sh
katya@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ echo 'date >> output.log' >> ~/date_script.sh
katya@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ chmod +x ~/date_script.sh
katya@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ cat > ~/Dockerfile << 'EOF'
> FROM ubuntu:latest
RUN apt-get update && apt-get install -y nano
COPY date_script.sh /date_script.sh
RUN chmod +x /date_script.sh
CMD ["/bin/bash"]
EOF
```

3. Собрать образ:

```
katya@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ docker build -t my_image ~/
[+] Building 18.1s (10/10) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.0s
=> => transferring dockerfile: 187B                                0.0s
=> [internal] load metadata for docker.io/library/ubuntu:latest   2.2s
=> [auth] library/ubuntu:pull token for registry-1.docker.io      0.0s
=> [internal] load .dockerignore                                   0.0s
=> => transferring context: 2B                                       0.0s
=> [1/4] FROM docker.io/library/ubuntu:latest@sha256:186072bba1b2f 4.0s
=> => resolve docker.io/library/ubuntu:latest@sha256:186072bba1b2f 0.0s
=> => sha256:817807f3c64e0b90b66edc7d90297f121ca 29.73MB / 29.73MB 3.0s
=> => extracting sha256:817807f3c64e0b90b66edc7d90297f121cad2a7c2a 0.9s
=> [internal] load build context                                   0.1s
=> => transferring context: 72B                                       0.0s
=> [2/4] RUN apt-get update && apt-get install -y nano             7.9s
=> [3/4] COPY date_script.sh /date_script.sh                      0.0s
=> [4/4] RUN chmod +x /date_script.sh                             0.3s
=> exporting to image                                              3.4s
=> => exporting layers                                              2.6s
=> => exporting manifest sha256:4857b2b2bf4579e25e7a2f54c40b046dd7 0.0s
=> => exporting config sha256:c8061ab555f6e30f883e7f9e0edc36151054 0.0s
=> => exporting attestation manifest sha256:f5b885aeea73b1786b0551 0.0s
=> => exporting manifest list sha256:30426869884cb6891d8707b10c8d9 0.0s
=> => naming to docker.io/library/my_image:latest                  0.0s
=> => unpacking to docker.io/library/my_image:latest                0.7s
```

4. Запустить контейнер:

```
katya@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ docker run -it --name my_container my_image /bin/bash
root@0757c1f8dd25:/#
root@0757c1f8dd25:/# /date_script.sh
root@0757c1f8dd25:/# cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
```

5. Внутри контейнера выполнить скрипт:

```
root@0757c1f8dd25:/# /date_script.sh
root@0757c1f8dd25:/# cat /output.log
Fri Mar 20 20:17:51 UTC 2026
Sat Mar 21 07:59:33 UTC 2026
Sat Mar 21 08:20:34 UTC 2026
```

6. Вывести список пользователей в образе:

```
root@0757c1f8dd25:/# cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534:./nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
ubuntu:x:1000:1000:Ubuntu:/home/ubuntu:/bin/bash
root@0757c1f8dd25:/# |
```

3. Git (GitHub)

1. Создать репозиторий на GitHub.
2. Создать структуру репозитория:

```
katya@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ git config --global user.name "kkatya05"
katya@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ git config --global user.email "katya.koliada@mail.ru"
katya@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ cd "C:\Users\katya\Desktop\C++ Labs"
-bash: cd: C:\Users\katya\Desktop\C++ Labs: No such file or directory
katya@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ cd 'C:\Users\katya\Desktop\C++ Labs'
-bash: cd: C:\Users\katya\Desktop\C++ Labs: No such file or directory
katya@LAPTOP-MK7SQ0DI:/mnt/c/Users/katya$ cd ~
katya@LAPTOP-MK7SQ0DI:~$ mkdir 'C++ Labs'
katya@LAPTOP-MK7SQ0DI:~$ cd 'C++ Labs'
katya@LAPTOP-MK7SQ0DI:~/C++ Labs$ mkdir -p lab_01/build lab_01/src lab_01/doc lab_01/cmake
katya@LAPTOP-MK7SQ0DI:~/C++ Labs$ mkdir -p lab_02/build lab_02/src lab_02/doc lab_02/cmake
katya@LAPTOP-MK7SQ0DI:~/C++ Labs$ ls -R
.:
lab_01  lab_02

./lab_01:
build  cmake  doc  src

./lab_01/build:

./lab_01/cmake:

./lab_01/doc:

./lab_01/src:

./lab_02:
build  cmake  doc  src

./lab_02/build:

./lab_02/cmake:

./lab_02/doc:

./lab_02/src:
katya@LAPTOP-MK7SQ0DI:~/C++ Labs$ echo "# C++ Labs" > README.md
katya@LAPTOP-MK7SQ0DI:~/C++ Labs$ echo "Лабораторные по C++" >> README.md
katya@LAPTOP-MK7SQ0DI:~/C++ Labs$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
```

```

hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/katya/C++ Labs/.git/
katya@LAPTOP-MK7SQ0DI:~/C++ Labs$ git add .
katya@LAPTOP-MK7SQ0DI:~/C++ Labs$ git commit -m "Initial commit: lab structure"
[master (root-commit) 3ad9d1b] Initial commit: lab structure
 1 file changed, 2 insertions(+)
 create mode 100644 README.md
katya@LAPTOP-MK7SQ0DI:~/C++ Labs$ git remote add origin https://github.com/kkatya05/C-Labs.git
katya@LAPTOP-MK7SQ0DI:~/C++ Labs$ git branch -M main
katya@LAPTOP-MK7SQ0DI:~/C++ Labs$ git push -u origin main
Username for 'https://github.com': github_pat_11CALOI0Q0eicxHRQHNdYr_YfTXdKcFK3bqXhPaIqcLH8^[[D^[[D^[[^C
katya@LAPTOP-MK7SQ0DI:~/C++ Labs$ git push -u origin main
Username for 'https://github.com': kkatya05
Password for 'https://kkatya05@github.com':
remote: invalid credentials
fatal: Authentication failed for 'https://github.com/kkatya05/C-Labs.git/'
katya@LAPTOP-MK7SQ0DI:~/C++ Labs$ git push -u origin main
Username for 'https://github.com': kkatya05
Password for 'https://kkatya05@github.com':
katya@LAPTOP-MK7SQ0DI:~/C++ Labs$ git push -u origin main
Username for 'https://github.com': kkatya05
Password for 'https://kkatya05@github.com':
remote: Invalid username or token. Password authentication is not supported for Git operations.
fatal: Authentication failed for 'https://github.com/kkatya05/C-Labs.git/'
katya@LAPTOP-MK7SQ0DI:~/C++ Labs$ git push -u origin main
Username for 'https://github.com': kkatya05
Password for 'https://kkatya05@github.com':
remote: Permission to kkatya05/C-Labs.git denied to kkatya05.
fatal: unable to access 'https://github.com/kkatya05/C-Labs.git/': The requested URL returned error: 403
katya@LAPTOP-MK7SQ0DI:~/C++ Labs$ git remote -v
origin  https://github.com/kkatya05/C-Labs.git (fetch)
origin  https://github.com/kkatya05/C-Labs.git (push)
katya@LAPTOP-MK7SQ0DI:~/C++ Labs$ git push -u origin main
Username for 'https://github.com': kkatya05
Password for 'https://kkatya05@github.com':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 265 bytes | 88.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/kkatya05/C-Labs.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

```

3. Создать ветки dev, stg, prd, удалить ранее существовавшие (локально и удалённо).

```

katya@LAPTOP-MK7SQ0DI:~/C++ Labs$ git checkout -b dev
Switched to a new branch 'dev'
katya@LAPTOP-MK7SQ0DI:~/C++ Labs$ git checkout -b stg
Switched to a new branch 'stg'
katya@LAPTOP-MK7SQ0DI:~/C++ Labs$ git checkout -b prd
Switched to a new branch 'prd'
katya@LAPTOP-MK7SQ0DI:~/C++ Labs$ git checkout dev
Switched to branch 'dev'
katya@LAPTOP-MK7SQ0DI:~/C++ Labs$ git push origin dev
Username for 'https://github.com': kkatya05
Password for 'https://kkatya05@github.com':
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0
remote:
remote: Create a pull request for 'dev' on GitHub by visiting:
remote:   https://github.com/kkatya05/C-Labs/pull/new/dev
remote:
To https://github.com/kkatya05/C-Labs.git
 * [new branch]      dev -> dev
katya@LAPTOP-MK7SQ0DI:~/C++ Labs$ git push origin stg
Username for 'https://github.com': kkatya05
Password for 'https://kkatya05@github.com':
remote: Invalid username or token. Password authentication is not supported for Git operations.
fatal: Authentication failed for 'https://github.com/kkatya05/C-Labs.git/'
katya@LAPTOP-MK7SQ0DI:~/C++ Labs$ git push origin stg
Username for 'https://github.com': kkatya05
Password for 'https://kkatya05@github.com':
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0
remote:
remote: Create a pull request for 'stg' on GitHub by visiting:
remote:   https://github.com/kkatya05/C-Labs/pull/new/stg
remote:
To https://github.com/kkatya05/C-Labs.git
 * [new branch]      stg -> stg
katya@LAPTOP-MK7SQ0DI:~/C++ Labs$ git push origin prd
Username for 'https://github.com': kkatya05
Password for 'https://kkatya05@github.com':
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0
remote:
remote: Create a pull request for 'prd' on GitHub by visiting:
remote:   https://github.com/kkatya05/C-Labs/pull/new/prd
remote:
To https://github.com/kkatya05/C-Labs.git
 * [new branch]      prd -> prd
katya@LAPTOP-MK7SQ0DI:~/C++ Labs$ git branch -d main
Deleted branch main (was 3ad9d1b).
katya@LAPTOP-MK7SQ0DI:~/C++ Labs$ git push origin --delete main

```

4. Создать скрипт merge_dev_to_stg.sh в корне репозитория для автоматического переноса ревизий из dev в stg с установкой тега (дата+время).

Сделать скрипт исполняемым: `chmod +x merge_dev_to_stg.sh`


```
katya@LAPTOP-MK7SQ0DI:~/C++ Labs$ cat > merge_dev_to_stg.sh << 'EOF'
> #!/bin/bash
> git checkout stg
> git pull origin stg
> git merge dev --no-ff
> TAG_NAME="stg-$(date +%Y%m%d-%H%M%S)"
> git tag "$TAG_NAME"
> git push origin stg
> git push origin "$TAG_NAME"
> git checkout dev
> echo "Перенос из dev в stg выполнен. Тер: $TAG_NAME"
> EOF
katya@LAPTOP-MK7SQ0DI:~/C++ Labs$ chmod +x merge_dev_to_stg.sh
```

5. Создать скрипт `merge_stg_to_prd.sh` в корне репозитория для автоматического переноса ревизий из `stg` в `prd` с установкой тега (дата+время).

```
katya@LAPTOP-MK7SQ0DI:~/C++ Labs$ cat > merge_stg_to_prd.sh << 'EOF'
> #!/bin/bash
> git checkout prd
> git pull origin prd
> git merge stg --no-ff
> TAG_NAME="prd-$(date +%Y%m%d-%H%M%S)"
> git tag "$TAG_NAME"
> git checkout dev
> echo "Перенос из stg в prd выполнен. Тер: $TAG_NAME"
> EOF
katya@LAPTOP-MK7SQ0DI:~/C++ Labs$ chmod +x merge_stg_to_prd.sh
katya@LAPTOP-MK7SQ0DI:~/C++ Labs$ git add merge_dev_to_stg.sh merge_stg_to_prd.sh
```

```
katya@LAPTOP-MK7SQ0DI:~/C++ Labs$ ./merge_dev_to_stg.sh
Switched to branch 'stg'
From https://github.com/kkatya05/C-Labs
 * branch          stg          -> FETCH_HEAD
Already up to date.
Merge made by the 'ort' strategy.
 merge_dev_to_stg.sh | 10 ++++++++
 merge_stg_to_prd.sh | 8 ++++++
 2 files changed, 18 insertions(+)
 create mode 100755 merge_dev_to_stg.sh
 create mode 100755 merge_stg_to_prd.sh
Username for 'https://github.com': kkatya05
Password for 'https://kkatya05@github.com':
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 227 bytes | 227.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/kkatya05/C-Labs.git
 3ad9d1b..f646c23 stg -> stg
Username for 'https://github.com': kkatya05
Password for 'https://kkatya05@github.com':
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/kkatya05/C-Labs.git
 * [new tag]          stg-20260321-171535 -> stg-20260321-171535
Switched to branch 'dev'
Перенос из dev в stg выполнен. Тер: stg-20260321-171535
katya@LAPTOP-MK7SQ0DI:~/C++ Labs$ ./merge_stg_to_prd.sh
Switched to branch 'prd'
From https://github.com/kkatya05/C-Labs
 * branch          prd          -> FETCH_HEAD
Already up to date.
Merge made by the 'ort' strategy.
 merge_dev_to_stg.sh | 10 ++++++++
 merge_stg_to_prd.sh | 8 ++++++
 2 files changed, 18 insertions(+)
 create mode 100755 merge_dev_to_stg.sh
 create mode 100755 merge_stg_to_prd.sh
Switched to branch 'dev'
Перенос из stg в prd выполнен. Тер: prd-20260321-171650
```

Вывод

В ходе лабораторной работы были освоены:

- управление пользователями и группами в UNIX-подобной ОС;
- настройка прав доступа с использованием `chmod` и `setfacl`;
- работа с Docker (сборка образов, запуск контейнеров);
- основы работы с Git (ветвление, слияние, теги);
- автоматизация переноса изменений между ветками с помощью скриптов.

Все задачи выполнены в среде Windows с использованием WSL2, что позволило получить полноценный UNIX-опыт без установки отдельной виртуальной машины.